

Graded Exercise 3: Analysis of Bacterial Genome Annotations

Objective

Development of a Python script for reading and systematically analyzing bacterial annotations provided in GFF3 and Fasta format.

Core Functionalities

1. Analysis of Sequence Features

- **Basic Metrics:**
 - Number of CDS
 - Number of RNAs (categorized by type)
- **Annotation Check:**
 - Number of CDS with ‘hypothetical protein’
 - Number of CDS with annotated ‘gene’
 - Number of CDS with COG categories.
- **Expansion:**
 - Identification of other existing features

2. Tabular Overview (CSV/TSV Export)

- Creation of a structured overview with the following columns:
 - ID (from the GFF file)
 - Feature type
 - Start position
 - Stop position
 - Gene name
 - Product
 - Strand

Note: Missing fields (e.g., gene name) should remain empty in the file

3. Sequence Extraction

- Extraction of all CDS sequences
- Storage as nucleotide and amino acid sequences
 - each as a Multiple Fasta file

4. Genome Analyses

- **Codon Usage:**
 - Calculation, averaged over all CDS
 - Analysis of the start codons used
- **COG Categories:**
 - Detailed evaluation of COG categories

5. Visualization

- **Creation of plots**
 - Develop at least two different meaningful visualizations of the data
 - e.g., bar charts or pie charts for COG distribution, CDS length distributions, codon usage plots ... be creative
 - Visualizations should be saved as image files

Technical Requirements

- **Command Line Interface:** Use of `argparse` for clean user guidance. Interactive modes (`input()`) or GUIs are only permitted as optional extensions
- **Programming Style:**
 - Strict modularization (no global variables, use of `main()`)
 - Meaningful and consistent variable names
 - Detailed documentation via comments

Optional Bonus Features

- **Comparative Analysis:** Analysis of multiple datasets including comparative metrics and plots
- **Filter Logic:** Export functions based on criteria such as gene length, presence of COG categories, or exclusion of ‘hypotheticals’
- **Genomic Checks:** Identification of overlapping genes, search for pseudogenes, or adjacent similar annotations
- **k-mer Analysis:** Calculation of the distribution of k-mers for different ‘k’
- **NCBI Integration:** Comparison with external data via accession numbers (using Biopython) and generation of RefSeq links
- **Advanced Reports:** Creation of reports in formats such as HTML, PDF, or Excel
- **Alternative Start Codons:** Analysis of upstream/downstream regions